DIPOLE TYPE BY LENGTH

Compute first *$\lambda = c/f$, then ratio l/λ*

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}}=199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\,\frac{e^{-jkR}}{R}\sin\theta,\quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta = S_{\max}\sin^2\theta\ (\text{W/m}^2)$$

$S(R,\theta)$ — avg. pwr dens. at (R,θ) (W/m^2)
l — antenna length (m, the rod itself)
I_0 — peak current in dipole (A)
R — distance from antenna to obs. point (m)
θ — angle from dipole axis (rad/°)
$k = 2\pi/\lambda$ — wavenumber (rad/m)
$\eta_0 = 120\pi \approx 377\,\Omega$ — free-space impedance
Max direction *broadside* $\theta_{\max} = \pi/2$
Total radiated power *integrate over 4π*

$$P_{\text{rad}} = \frac{\eta_0\pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2\ (\text{W})$$

Directivity *exact* $D_{\text{Hertz}} = 1.5$ (1.76 dB)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow$ $S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\to$ tiny $\to 0$.

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

Quarter-wave $l = \lambda/4$ *$\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$*

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2,\quad S_0 = \frac{15 I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\,S_0$$

Half-wave $l = \lambda/2$ *$\pi l/\lambda = \pi/2$, shortcut form*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\,\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$$

$D = 1.64$, $R_{\text{rad}} = 73\,\Omega$, $\theta_{\max} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180}\qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

°	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max}\ (\text{unitless})$$

Step 2 — HPBW β *half-power beamwidth, full angular width at -3 dB*

$$\text{Set } F(\theta) = 0.5,\ \text{solve for } \theta \ \to\ \theta_{1/2}$$
$$\beta = 2\theta_{1/2}\ (\text{symmetric beams})$$

***Report HPBW in degrees**, not rad. Multiply by $180/\pi$ at the end. Gaussian:*

$$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5\ \Rightarrow\ \theta_{1/2} = \sqrt{\ln 2/a}$$

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\,d\theta\,d\phi$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\,d\theta$.

TI-36X Pro: `[2nd][d/dx]` for $\int_\square^\square$, set RAD mode, multiply result by 2π.

Narrow-beam Gaussian shortcut ($\sin\theta \approx \theta$, $\to\infty$):

$$\Omega_p \approx \frac{\pi}{a} = \frac{\pi\,\theta_{1/2}^2}{\ln 2}\qquad (F = e^{-a\theta^2})$$

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p}\ \ (\text{unitless})\qquad D_{\text{dB}} = 10\log_{10} D$$

Alt. with two orthogonal HPBWs: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.

Step 5 — Gain (if asked) *ξ = rad. eff., $0 < \xi \le 1$*

$$G = \xi D\qquad G_{\text{dB}} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow$ $G = D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$ monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

Lossless antenna ($\xi = 1$) $\Rightarrow$ $G = D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G = 1.64$).

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi}\ (\text{m}^2)$$

Physical cross section *wire dipole, diameter d*

$$A_p = l\,d = \frac{\lambda}{2}\,d\ (\text{half-wave})$$

Inverse: D **from** A_e

$$D = \frac{4\pi A_e}{\lambda^2}\ (\text{unitless})$$

Hertzian special case *$D = 1.5$*

$$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2\ (\text{m}^2)$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$.

R_{rad}, EFFICIENCY, POWER SPLIT

Radiation resistance *Ω, equiv. ckt. load absorbing P_{rad}*

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2}\qquad R_{\text{loss}} = \frac{2P_{\text{loss}}}{I_0^2}$$

Input power split *P_t from generator*

$$P_t = P_{\text{rad}} + P_{\text{loss}}\qquad R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$$

Radiation efficiency *two equivalent forms*

$$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}\ (\text{unitless})$$

Receiving antenna *$\tilde{V}_{oc} = \tilde{E}^i l$ for short dipole*

Max pwr transfer when $Z_L = Z_{\text{in}}^*$ (conjugate match):

$$P_L = P_{\text{int}} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}\ (\text{W})$$

$$A_e = \frac{30\pi|\tilde{V}_{\text{oc}}|^2}{R_{\text{rad}}|\tilde{E}^i|^2}\ (\text{m}^2,\ \text{alt. form})$$

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr *G_t, G_r linear gains; $\lambda = c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2}\ (\text{W/m}^2);\quad \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t}\ (\text{W})$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}}\ (\text{m})$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2},\quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t/L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t\,F_t(\theta_t,\phi_t)F_r(\theta_r,\phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.
3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t, SNR, etc.

$$X_{\text{dB}} = 10\log_{10} X_{\text{lin}}\qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g. $D_{dB} = 10\log_{10} D$; $G_{dB} = 10\log_{10} G$; $SNR_{dB} = 10\log_{10}(P_r/P_N)$.

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

POWER FLOW IDENTITIES

$$P_{\text{rad}} = S_{\max}\,\Omega_p\,R^2 = \frac{4\pi R^2 S_{\max}}{D}\ (\text{W})$$

$$S_{\max} = \frac{P_{\text{rad}}\,D}{4\pi R^2}\ (\text{W/m}^2)$$

Isotropic radiator: $S = P_{rad}/(4\pi R^2)$ (no D factor).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B\,T_{\text{sys}}\,B\ (\text{W})$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N}\ (\text{unitless})\qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2}\ (\text{unitless})\quad\Rightarrow\ A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}\ (\text{use rad});\ \text{circular: } D \approx 4\pi/\beta^2$$

HPBW and D **by aperture shape** *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x\ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$, $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x\beta_y) = 4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$

Scaling rules *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

Change	New D	New β
Double area ($A_p \to 2A_p$)	$D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda \to \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$

LINEAR ANTENNA ARRAY

Setup *N identical elements, spacing d along axis*

element pos $z_i = id,\ i = 0, \ldots, N-1;\ A_i = a_i e^{j\psi_i}$

Pattern multiplication

$$S(R, \theta, \phi) = S_e(R, \theta, \phi)\, F_a(\theta)$$

Array factor $F_a(\theta) = |\sum A_i e^{jkid\cos\theta}|^2$

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i\, e^{jkid\cos\theta} \right|^2$$

Uniform $A_i = 1$, equal phase *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)},\ \gamma = kd\cos\theta$$

Normalize: F_a/N^2. Peak N^2 at $\gamma = 0$ ($\cos\theta = 0$, broadside).

HPBW (broadside, uniform) *array length $L = (N-1)d$*

$$\beta \approx 0.88\,\frac{\lambda}{L}\ \text{(rad)}$$

SYMBOL REFERENCE

l — antenna length (m); $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
I_0 — peak current (A); $\eta_0 = 120\pi \approx 377\,\Omega$
$k = 2\pi/\lambda$ wavenumber (rad/m); R obs. dist (m)
$S(R, \theta)$ — power density (W/m^2)
$F(\theta) = S/S_{\max}$ — normalized pattern (unitless)
β — half-power beamwidth (rad or deg)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity (unitless); ξ rad. eff.; $G = \xi D$
$A_e = \lambda^2 D/(4\pi)$ — effective area (m^2); $A_p = ld$ phys. (m^2)
R_{rad} — radiation resistance (Ω); R_{loss} loss res. (Ω)
P_t, P_r — TX / RX power (W); G_t, G_r linear gains